EEE 206

PID vs FUZZY

Controllers in PMSM motor

BY: GROUP 02

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Problem Statement

In PMSM Motor the PID controller shows reduced performance under varying loads. Our aim is to design a fuzzy logic controller to overcome this drawback, and compare the disturbance rejection of both the controllers.

- What is PID?
- What is Fuzzy?
- What is PMSM Motor?

PMSM, PID And Fuzzy

PMSM MOTOR

The define feature is that the rotor's magnetic field is provided by permanent magnets, rather than by rotor current or field windings like in any other induction motor.

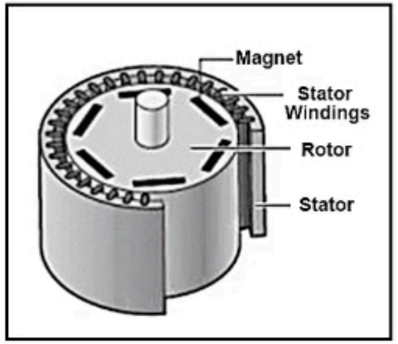
PID Controllers

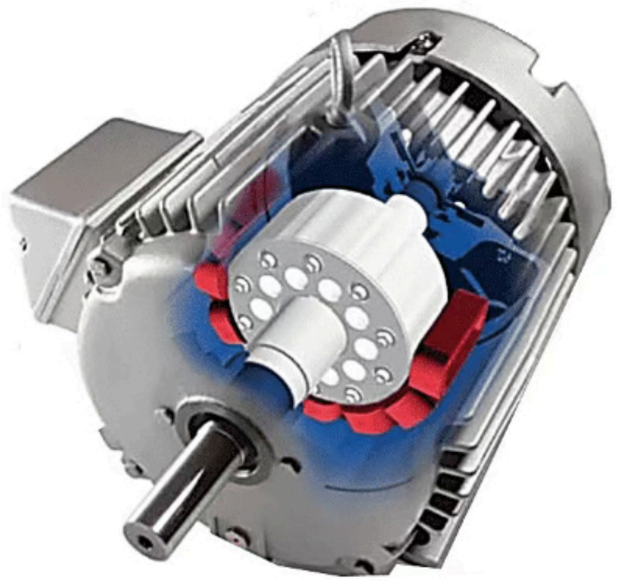
A PID controller (**Proportional–Integral–Derivative controller**) is a control system used to maintain a desired output by minimizing error.

Fuzzy Controllers

A fuzzy controller is a control system based on Fuzzy Logic, It uses if–then rules (like human thinking) to make decisions, rather than precise fixed mathematical models.

PMSM Mechanical Equation





PMSM Mechanical Equation (With Load)

When load torque is considered, the equation becomes:

$$T_e - T_L = J\,(d\omega/dt) + B\omega$$

Taking Laplace transform:

$$T_e(s) - T_L(s) = (Js + B)\,\omega(s)$$

Rearranging:

$$\omega(s) = [T_e(s) - T_L(s)] / (Js + B)$$

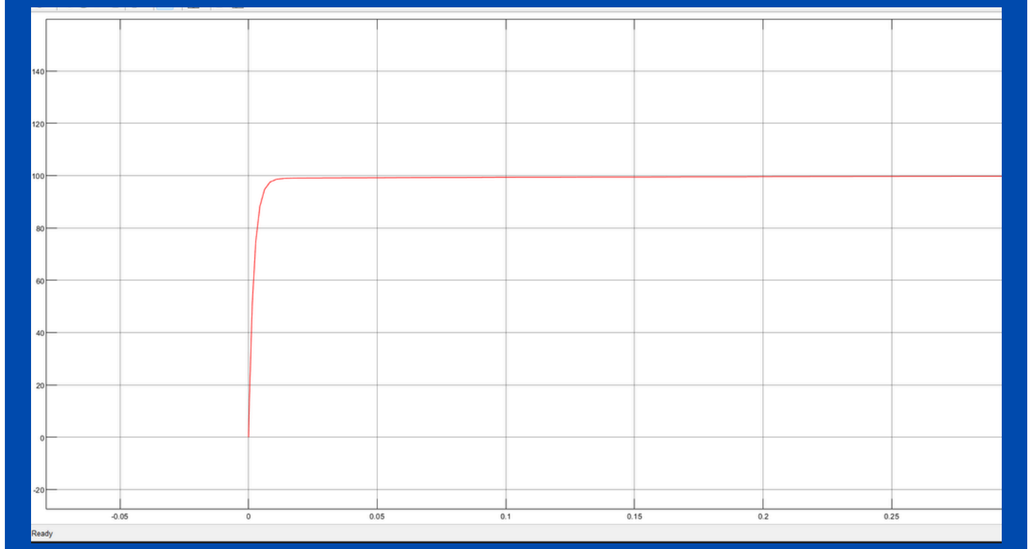
Thus, the transfer function is:

$$G(s) = \omega(s)/T_e(s) = 1 / (Js + B)$$

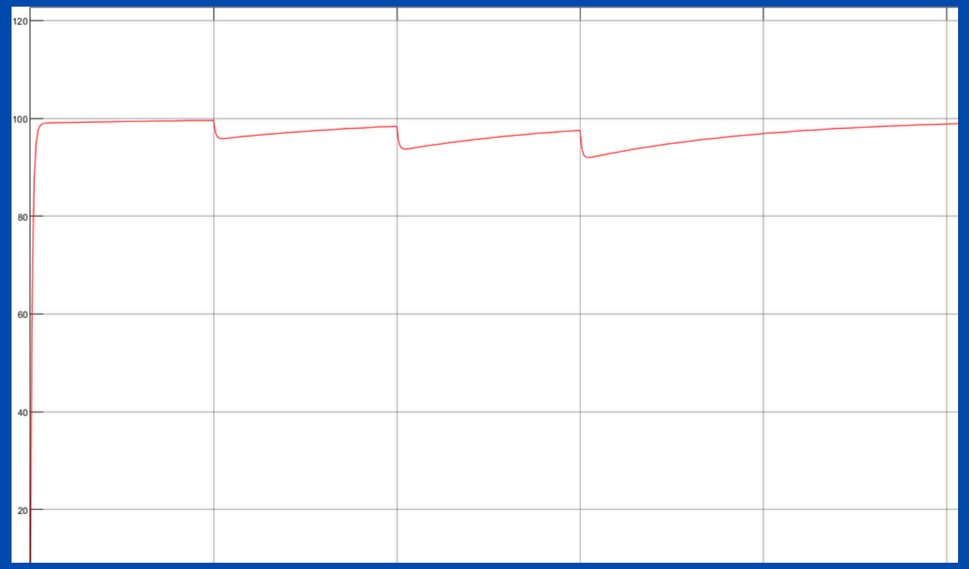
The load torque acts as a disturbance input:

$$G_d(s) = \omega(s)/T_L(s) = -1 / (Js + B)$$

PID Controller Simulation



PID Controller Simulation of PMSM without any loads

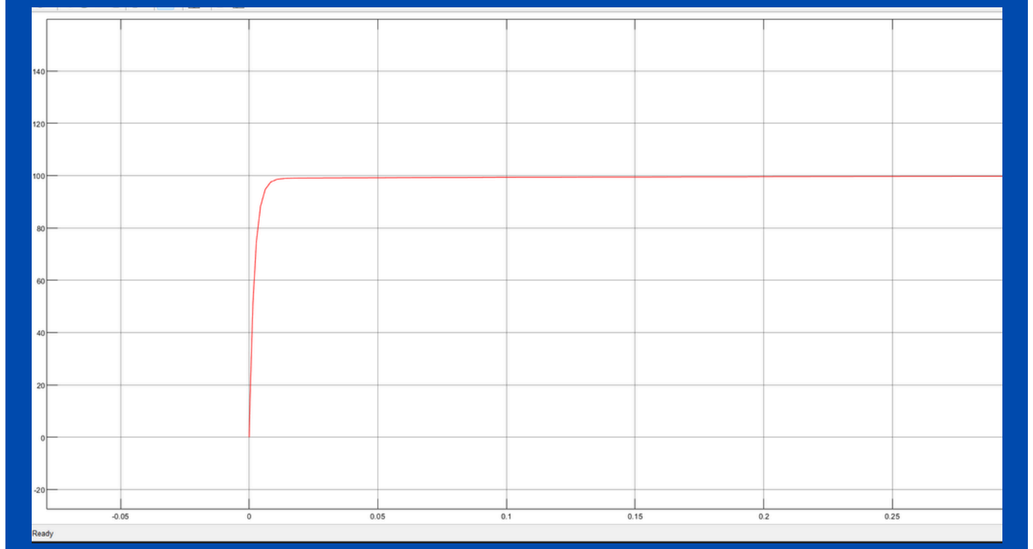


PID Controller Simulation of PMSM with loads

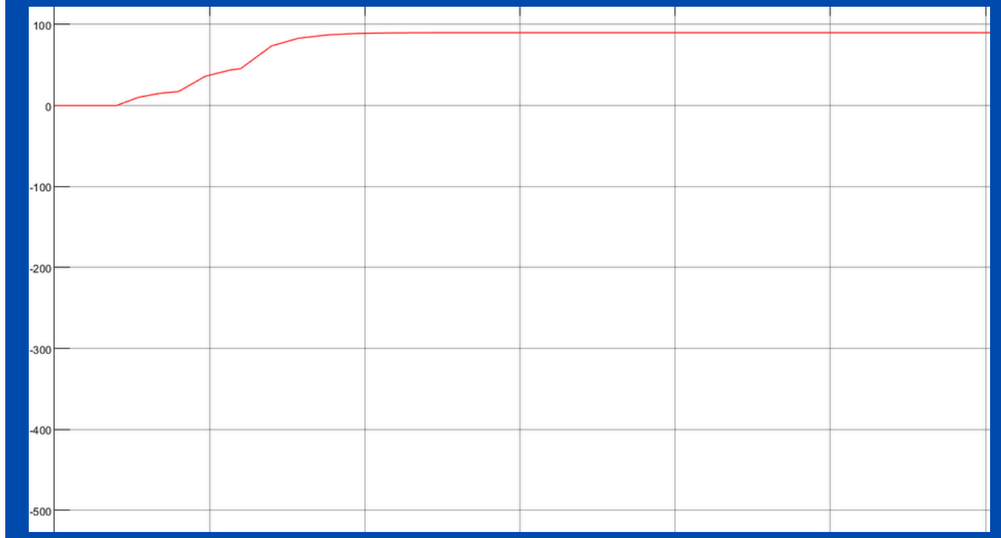
PID Controller Limitations

- Performance decreases for nonlinear systems like motor drives under changing conditions.
- Requires manual tuning of Kp, Ki, and Kd, which can be time-consuming
- May produce overshoot and oscillations if not tuned properly. Cannot adapt automatically to parameter variations in the system.

Fuzzy Controller Simulation



Fuzzy Controller Simulation of PMSM without any loads



Fuzzy Controller Simulation of PMSM with loads

PID Controller vs Fuzzy Controller in PMSM

Disturbance Rejection Performance of PID

1. Minimum Speed dip after Disturbance

Speed dip = 19 rad/s

2. Recovary time after applying 3 loads at 3 different times -2% to +2% tolerance

TR= 1.25-0.19= 1.065s

Disturbance Rejection Performance of Fuzzy

1. Minimum Speed dip after Disturbance

Speed dip =2 rad/s

2. Recovary time after applying 3 loads at 3 different times -2% to +2% tolerance

TR= 0.0.43-0.041 = 0.002s

Conclusion

- A comparative analysis of PID and Fuzzy Logic Controllers for the speed control of a Permanent Magnet Synchronous Motor (PMSM) has been presented.

- The system was modeled and simulated in MATLAB/Simulink, and the performance of both controllers was evaluated under different operating conditions.

- Therefore, it can be concluded that fuzzy logic control is a more effective and reliable approach for PMSM speed control in applications involving nonlinearities and varying load conditions.

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Thank You

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